

Securing Web Applications: A Machine Learning Approach for SQL Injection Threats

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Abstract

The rapid evolution and widespread adoption of the internet have significantly transformed the world, leading to an increased number of cyber attacks. Cyber security has become one of the most critical challenges for society, incurring substantial financial losses annually. This research focuses on the specific threat of SQL Injection (SQLi) attacks on web applications, aiming to detect malicious queries designed to exploit vulnerabilities and access sensitive data.

In recent years, the frequency of SQLi attacks has surged, posing severe risks to web application security. Attackers use SQLi to execute arbitrary SQL code, potentially gaining unauthorized access to databases, exfiltrating data, and compromising the integrity of web services. Despite various efforts to mitigate SQLi attacks, a comprehensive and effective detection mechanism remains elusive.

This paper highlights the application of machine learning approach in identifying and preventing SQLi attacks, providing a comparative analysis of various classifiers and their performance. The study underscores the potential of machine learning in enhancing web application security, offering adaptive and dynamic solutions to combat the evolving threat of SQLi attacks.

Keywords-SQL Injection, Threat Detection, Web Security, Machine Learning.

I. Introduction

The field of cybersecurity has experienced tremendous growth, addressing a variety of threats, including SQL Injection (SQLi) attacks, which are among the most significant dangers in the realm of web applications. SQLi attacks represent a persistent threat to database-driven applications, exploiting vulnerabilities to execute malicious SQL code. These attacks are particularly challenging to prevent because they can be

easily executed by inserting malicious queries into input fields, causing immediate and severe damage.

SQL Injection attacks allow cybercriminals to compromise web applications, leading to unauthorized access to sensitive data, identity theft, and other malicious activities. Attackers manipulate SQL queries to bypass security measures, access backend databases, and perform actions such as data theft, deletion, or modification.

Specially crafted SQL queries are used by attackers to exploit these vulnerabilities, gaining control over the database and potentially the entire application. Cybercriminals target web applications to create extensive networks of compromised systems, which they can manage and exploit remotely.

Traditional SQLi detection techniques, such as signature-based and rule-based methods, may not be sufficient to detect modern and sophisticated SQLi threats. These conventional methods often struggle to keep pace with the evolving tactics employed by attackers. Therefore, there is a critical need for advanced detection mechanisms that can adapt to new attack patterns.[1]

This research paper investigates and explores a machine learning based approach for recognizing and mitigating SQL Injection threats, focusing on developing a robust detection system capable of identifying various attack vectors with high accuracy. By leveraging natural language processing and advanced classification algorithms, this study of research aims to provide an adaptive and dynamic solution for defending against SQL Injection threats in web applications.

II . Literature Survey :

This section discusses about the literature survey of different researchers in the field of virtualization and network security. The researcher Ganji et al. [2] examined suitable infrastructure for the linux OS. They also mentioned linux system security requirements in

server network security. Patil et al. [3] proposed hypervisor level distributed network security (HLDNS) framework which is deploy to monitor VM related network traffic for intrusion detection. Li et al. [4] proposed framework, which divides network security into five stages which are as factor acquisition, model representation, measurement establishment, solution analysis and situation prediction. Bock et al. [5] explain real time hypervisor based on Xvisor for delivering secure and separated environment for virtualized system. Compastie et al. [6] describe comparison between different virtualization models to build architecture of cloud infrastructure for analyzing network security issues. Yan et al. [7] verified effectiveness of system using client-side distributed energy storage demo project for improving system load characteristics. Potdar et al. [8] describes server virtualization, which provides a platform to run different OS services. They describe performance evaluation of docker container and virtual machines using effective tools. Bahn et al. [9] explains separations of VM for managing unnecessary network traffic.

III. Machine Learning Framework for SQL Injection Detection

In many studies, machine learning approach to detect SQL Injection threat has been extensively discussed. Various researchers have introduced diverse approaches to enhance the accuracy and performance of SQL Injection threat detection. The proposed SQL Injection detection technique used supervised machine learning algorithm and is intended to detect SQL Injection as mentioned in Fig.1. First collected SQL Injection dataset from Kaggle. The dataset contain malicious and benign query. Data preprocessing performed on dataset for cleaning data. Feature are extracted from dataset using three different method 1) CountVectorizer 2)TF-IDF 3) Word2Vec from Word Embeddings. The data from dataset is divided into 80% for training and 20% for testing. Model is then trained using SVM Classifier. Proposed model is tested and evaluated based on four different evaluation parameters such as Accuracy, Precision, Recall and F1-Measure for the three different ways of feature extraction and compared.

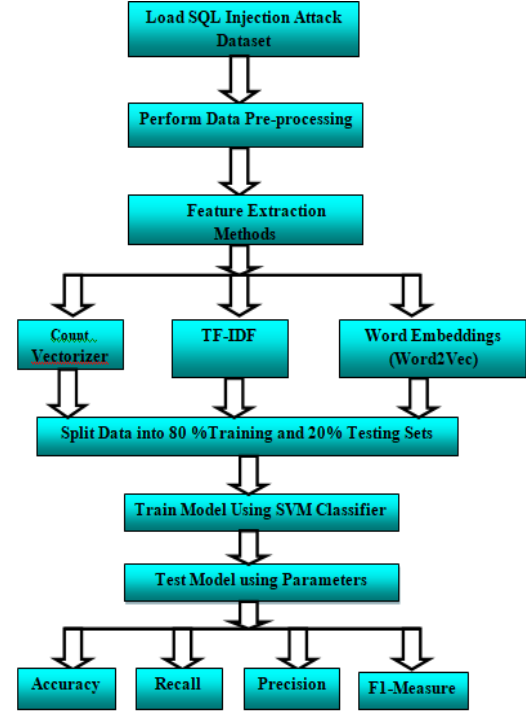


Fig.1. Proposed Approach for SQL Injection Attack Detection

IV. EVALUATION TERM DESCRIPTION

Sr.No	Evaluation Term	Description
1	Accuracy	Overall Performance Of The Model
2	Precision	Percentage of classified SQL Injection which are actually SQL Injection
3	Recall	Effectiveness of model in detecting SQL Injection
4	F-Measure	Blend of precision and recall

TABLE 1:DESCRIPTION OF EVALUATION TERMS

V. THE RESULTS AND DISCUSSION

The proposed technique is evaluated based on parameters Accuracy ,Precision, Recall and F1-Measure and shown in Table 2 .

Sr No	Feature Extraction Method	Evaluation Parameters			
		Accuracy	Precision	Recall	F-Measure
1	Count Vectorizer	0.9917	0.9898	0.9876	0.9887
2	TF-IDF	0.9765	0.9907	0.9449	0.9673
3	Word2Vec	0.9872	0.9897	0.9753	0.9824

TABLE 2: EVALUATION OF PROPOSED TECHNIQUES

VI. ANALYSIS AND RESULTS

In this work, the author M. Hasan et al. evaluated a heuristic machine learning approach that combines static and dynamic analysis. Author developed system using MATLAB utilized well established dataset and 23 machine learning classifiers for training and testing. The top five classifiers are selected based on their performance metrics, and a GUI application is developed to facilitate easy testing of new SQL statements[4]. The author T.P.Latchoumi et al. worked on machine learning algorithms, specifically the Support Vector Machine algorithm, to detect and prevent SQL injection threat. The SVM is trained with a dataset having both benign as well as malicious SQL queries. This training enables the SVM to generate a framework capable of distinguishing between legitimate and harmful queries. The system architecture includes a real-time monitoring component that applies the SVM model to incoming SQL queries to identify and block potential attacks[5]. The author Dr. Sandeep Kumar et al. used the "SQL1" and "SQL2" datasets from Kaggle, containing labeled SQL queries. Applied supervised learning algorithms such as Linear Discriminant Analysis (LDA), Logistic Regression (LR), and K-Nearest Neighbors (KNN). Implemented a stacking technique where predictions from base models (LDA, LR, KNN) are used as inputs to a meta model (LR) and highlights the potential of machine learning algorithms combined with stacking techniques to improve security[6]. Author B. Gogoi et al. developed Apache web server module that intercepts HTTP/HTTPS requests and inspects them for SQLi payloads. The system parses HTTP requests into tokens and converts them into numerical data using word embedding (word2vec). Synthetic Minority Oversampling Technique is used to handle imbalance of class in the dataset (SMOTE). Trains classifiers using Support Vector Machines (SVM) with both linear and non-linear models. Hyperparameters are optimized using Grid Search. Precision, F1-Score and recall parameters are used for the evaluation of system and achieved a high detection rate for all parameters to 99.9%. 10-Fold cross validation is used to prevent over fitting [7]. Author Prince Roy et al tested machine learning classifiers to detect payloads of SQL injection threat. The authors used SQL Injection dataset from the Kaggle for training and testing, a variety of classifiers, including Naïve Bayes, Logistic Regression, XGBoost, Naive Bayes, AdaBoost, and Random Forest[8]. Author J. M.Alkhathami et al. suggested the model which is trained on a dataset of SQL queries, both normal and injection attacks, collected from publicly available repositories. Data preprocessing involves cleaning the dataset and converting text values to numerical features

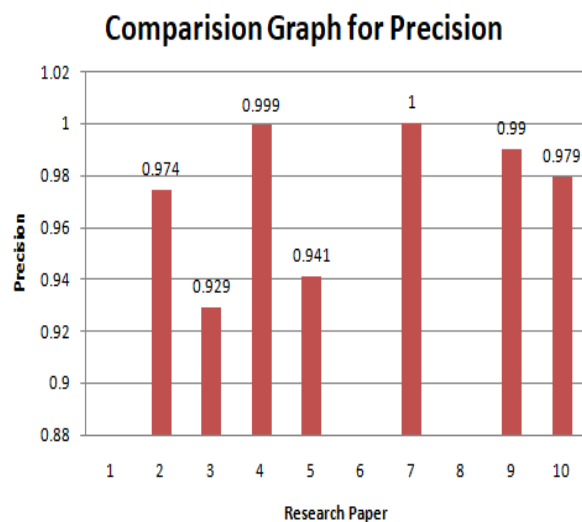
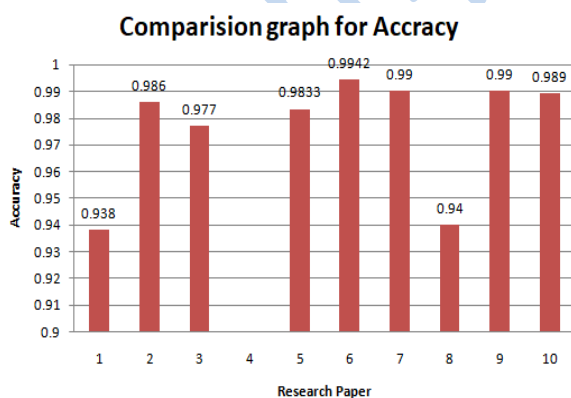
using the countVectorizer class. Four algorithms are used for training: KNN, MNB, DT, and SVM. The models are assessed using metrics such as accuracy, F-Value, confusion matrix and AUC-ROC curve. SVM achieved the maximum accuracy at 99.42%, after that DT at 99.4%, MNB at 97.09%, and KNN at 92.45%[9]. Author Ahmed Abadulla Ashlam et al. proposed methodology involves a arithmetical, systematic, and experimental approach to study the harmful effect of SQLi attacks. It includes the collection of a dataset from sources like the National Vulnerability Database and Lib-Injection. The methodology covers data preprocessing steps such as tokenization and stop words removal, feature extraction, and the development of a classification model using techniques like CountVectorizer and Support Vector Machine [10].

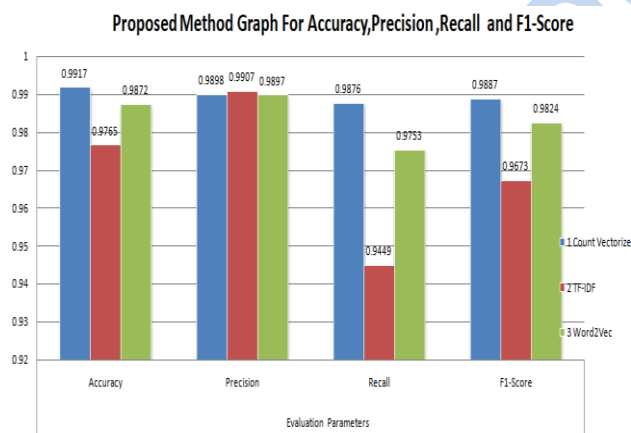
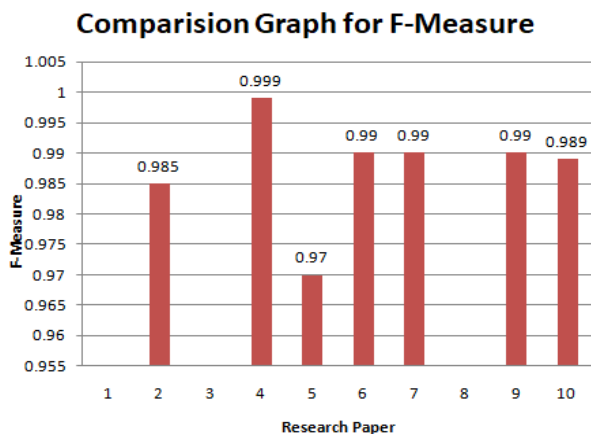
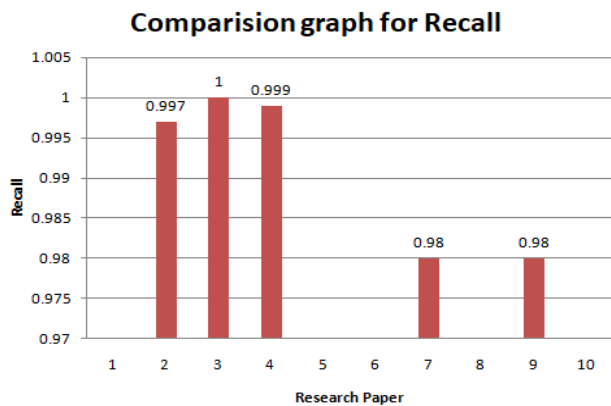
Author Yohan Muliono et al. employed Support Vector Machines (SVM) for grouping and predicting SQL Injection attacks. Focuses on SQL query conditions and program fragments vulnerable to attacks[11]. Author A. Zhumabekova et al. Worked on algorithms using machine learning, include Naïve Bayes, Decision Tree, Support Vector Machine, XGBoost, Random Forest and CatBoost. Used the dataset for experiments was sourced from Kaggle, containing 68,553 items divided into 70% for training and 30% for testing. The first step is data preprocessing, involve collecting, cleaning, normalizing, scaling, and vectorizing the data with the help of the tf-idf metric. The processed data is used to train and test a variety of machine learning models to assess their performance in detecting SQL injections. The results are assessed using parameters such as accuracy, F1-Score, precision and recall values[12]. Author Shengyuan Zhang et al. The developed a method, called the Feature Ratio Method, focuses on six main features 1) Length of payload 2) Symbol ratio 3) Comparison ratio 4) Payload complexity 5) Functional call ratio 6) Discontinuous semantics ratio. It uses regular expressions and fuzzy matching to extract these features and improve detection accuracy [13].

Sr. No	Name of Author	Classifier/Algorithm	Parameters			
			Accuracy	Precision	Recall	F-measure
1	M. Hasan et al.,2019 [2]	Ensemble Boosted Trees, Ensemble Bagged Trees, Linear Discriminant, Cubic SVM, Fine Gaussian SVM	0.938	-	-	-
2	T.P.Latchoumi et al.,2020 [3]	SVM	0.986	0.974	0.997	0.985
3	Dr. Sandeep Kumar et al.,2021 [4]	LR,KNN,LDA and Meta model	0.977	0.929	1.0	-
4	B. Gogoi et al.,2021 [5]	Linear SVM , Non linear SVM	-	0.999	0.999	0.999
5	P. Roy et al.,2022 [6]	Logistic Regression,AdaBoost, Random Forest,XGBoost,Naïve Bayes	0.9833	0.941	-	0.970
6	J. M. Alkhathami et al.,2022 [7]	KNN,MNB,Decision Tree,SVM	0.99	-	-	0.99
7	Ahmed Abadulla Ashlam et al.,2022 [8]	SVM, Decision Tree, KNeighbors, AdaBoost, Random Forest, SVM with PALOSDM	0.99	1	0.98	0.99
8	Yohan Muliono et al.,2022 [9]	SVM	0.94	-	-	-
9	A. Zhumabekova et al.,2023 [10]	Decision Tree, Random Forest, XGBoost, and CatBoost	0.99	0.99	0.98	0.99
10	Shengyuan Zhang et al.2023 [11]	LR,KNN,RF,SVM,NB, CNN	0.989	0.979	-	0.989

TABLE 3 :COMPARISION OF SQL INJECTION DETECTION PARAMETERS

VII.GRAPHICAL REPRESENTATION OF VARIOUS PARAMETRES





VII. CONCLUSION

This paper explores SQL Injection and presents detection technique using count vectorizer method for feature extraction with SVM classifier achieving an accuracy of 0.9917 which is best than TF-IDF method of feature extraction with SVM and Word2vec IDF method of feature extraction with SVM & Performance analysis of various parameters of existing methodology. Additionally, the study provides a detailed analysis of SQL Injection parameters, offering insights for researchers. The results are compared with the a variety of methods based on Performance metrics.

Competing Interest- Cyber security

Funding Information- Not Applicable

Author Contribution- After detailed analysis of SQL Injection attack detection techniques using machine learning author done the performance analysis of different feature extraction method and find out the best method which results in increasing the Accuracy of different machine learning algorithm.

Data Availability Statement- Dataset taken from Kaggle

Research involving human and /or Animals: This research is indirectly for human. As humans are going to use internet services and might have chances to get victim of SQL Injection Attack

Informed consent:- Not Applicable

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